

Mount Everest Accident Data Analysis Report

2020–2025

1. Executive Summary

This report presents an exploratory data analysis of 500 recorded Mount Everest accident records covering the period 2020–2025.

The objective of the analysis was to identify patterns in recorded accidents based on time, altitude, causes of death, season, weather conditions, experience level, age, gender, route, nationality, location and expedition company.

Several important patterns were identified. The most significant finding was the concentration of recorded incidents at extreme altitude. 344 of the 500 records (68.8%) occurred at 8,000 metres or above. Altitude sickness was the most frequently recorded cause, with 129 cases (25.8%), followed by exhaustion with 81 cases (16.2%).

The number of recorded accidents peaked at 93 in 2021 and declined to 70 in 2025. Summer contained the highest number of records at 193, while extreme cold was the most frequently recorded weather condition.

The dataset also shows an almost equal gender distribution, with 252 male and 248 female records.

However, the results should be interpreted as patterns within recorded accident data rather than accident risk rates. The dataset does not contain the total number of climbers, expeditions, route usage or exposure time. Consequently, it is not possible to determine the probability or rate of an accident for a particular group.

2. Project Objectives

The analysis aimed to answer the following questions:

1. How did recorded accidents change between 2020 and 2025?
2. What were the most common recorded causes of death?
3. At what altitudes did most recorded accidents occur?
4. How significant was the 8,000m+ Death Zone?
5. Which seasons had the most recorded accidents?
6. Which weather conditions appeared most frequently?

7. How were recorded accidents distributed by experience level?
 8. How were records distributed by gender and age?
 9. Which routes had the highest number of recorded accidents?
 10. Which nationalities and expedition companies had the highest recorded counts?
 11. What relationships can be observed between altitude and cause of death?
 12. What limitations should be considered when interpreting the results?
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3. Dataset Overview

The dataset contains:

- 500 records
- 13 original variables
- Dates covering 2020–2025
- Ages ranging from 20 to 69
- Recorded altitudes ranging from 5,364m to 8,848m
- 30 nationalities
- 6 routes
- 4 seasons
- 8 weather-condition categories
- 6 experience categories

- 16 expedition companies
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4. Data Cleaning

The dataset was inspected before performing the analysis.

The following checks and transformations were performed:

- Checked the number of rows and columns
- Inspected column names and data types
- Checked for missing values
- Checked for duplicate records
- Converted **Date** to datetime format
- Standardised text fields
- Converted numerical fields to numeric data types
- Validated age values
- Validated altitude values
- Created **Year**
- Created **Month**
- Created **Altitude_Band**
- Created **Age_Group**
- Created **Death_Zone**

Data Quality Results

Data Quality Check	Result
Total records	500
Original columns	13
Missing values	0
Duplicate records	0
Invalid dates	0
Minimum age	20
Maximum age	69
Minimum altitude	5,36 4m
Maximum altitude	8,84 8m

The dataset contained no missing values or duplicate rows, so no records needed to be removed for those reasons.

5. Annual Accident Trends

The number of recorded accidents varied throughout the six-year period.

Year	Recorded Accidents	YoY Change
2020	86	—
2021	93	+8.1%
2022	92	-1.1%
2023	86	-6.5%
2024	73	-15.1%
2025	70	-4.1%

Finding

The highest number of recorded accidents occurred in 2021 with 93 records.

After 2021, the number generally declined, reaching 70 records in 2025.

This represents a decrease of approximately 24.7% from 2021 to 2025.

Interpretation

The dataset indicates a downward trend in recorded accident counts after 2021.

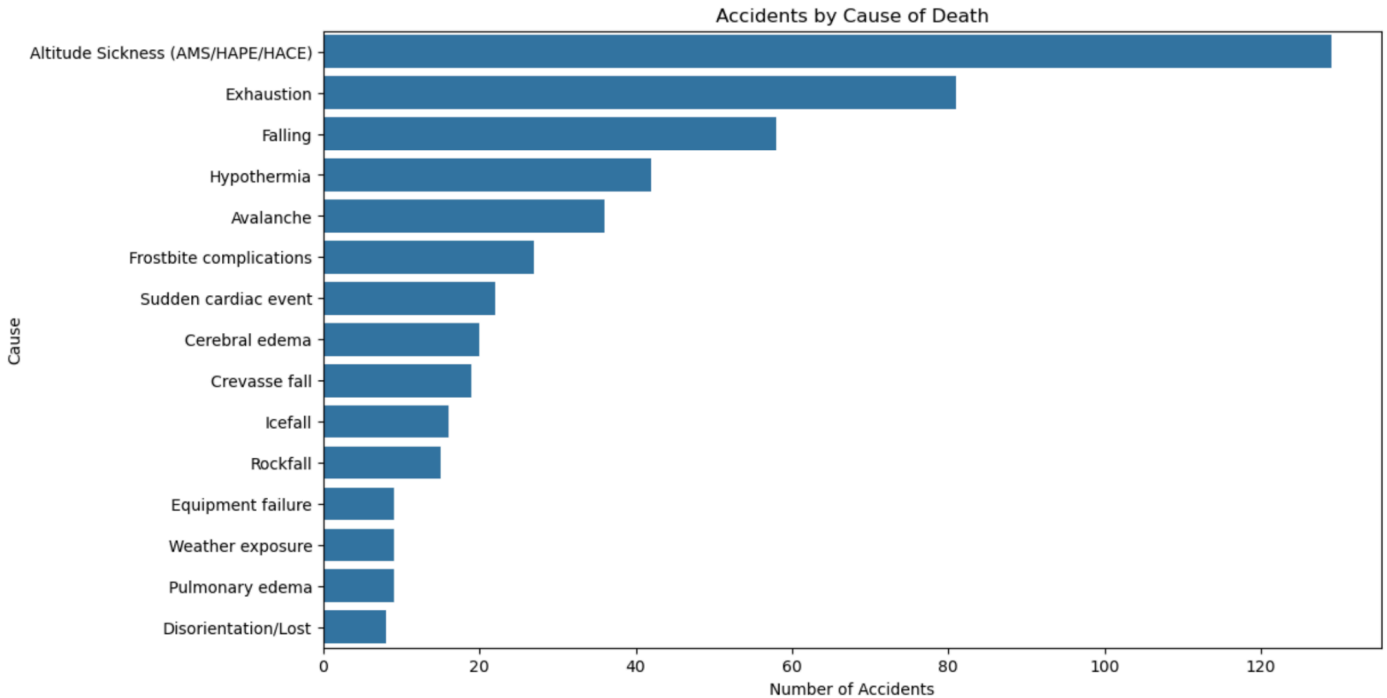
However, this should not automatically be interpreted as evidence that Everest became safer. The dataset does not contain information about the number of climbers or expeditions in each year.

6. Causes of Death

The cause-of-death analysis identified significant differences between categories.

Cause of Death	Records	Percentage %
Altitude Sickness (AMS/HAPE/HACE)	129	25.8%
Exhaustion	81	16.2%
Falling	58	11.6%
Hypothermia	42	8.4%
Avalanche	36	7.2%
Frostbite complications	27	5.4%
Sudden cardiac event	22	4.4%
Cerebral edema	20	4.0%
Crevasse fall	19	3.8%
Icefall	16	3.2%

The remaining causes accounted for smaller proportions of the dataset.



Key Finding

Altitude sickness was the leading recorded cause of death, accounting for 129 records or 25.8% of the dataset.

Exhaustion was the second-largest category at 81 records or 16.2%.

Together:

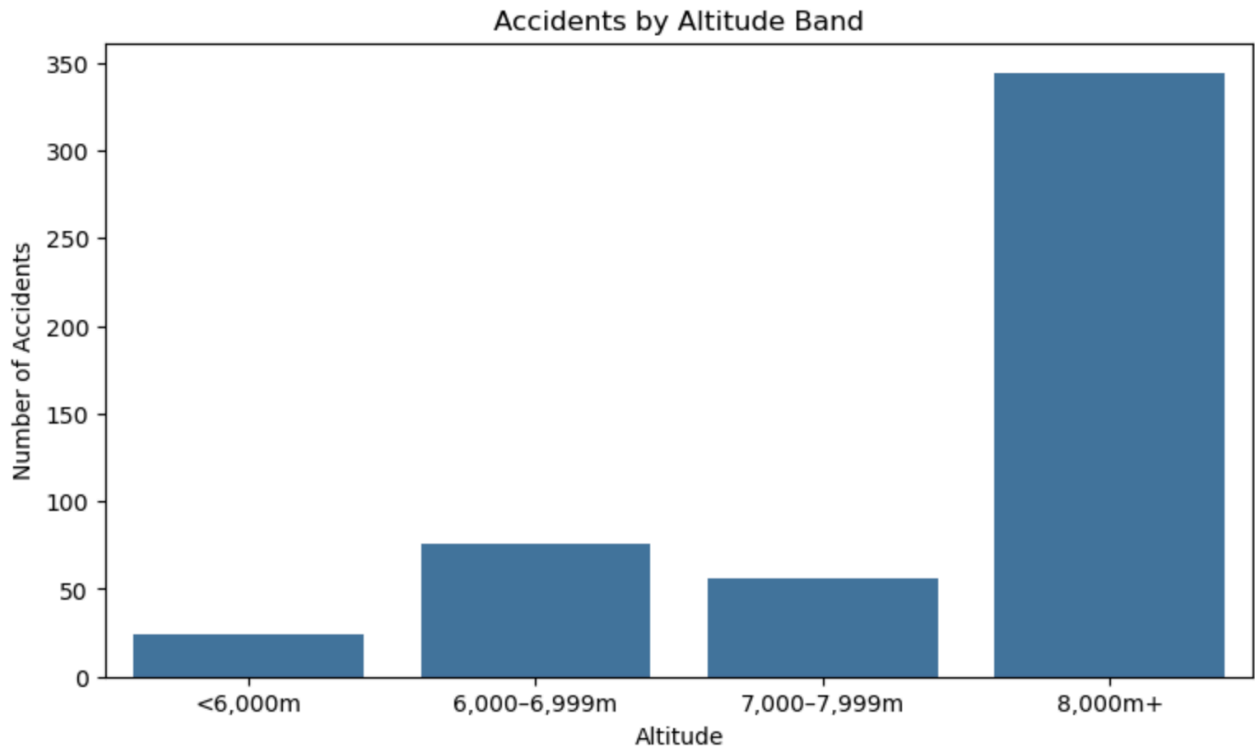
Altitude sickness + exhaustion = 210 records

This represents 42.0% of all records.

7. Altitude Analysis

Altitude was one of the strongest patterns identified in the analysis.

Altitude Band	Reco rds	Perce ntage
<6,000m	24	4.8%
6,000–6,999m	76	15.2%
7,000–7,999m	56	11.2%
8,000m+	344	68.8%



Key Finding

68.8% of all recorded accidents occurred at 8,000 metres or above.

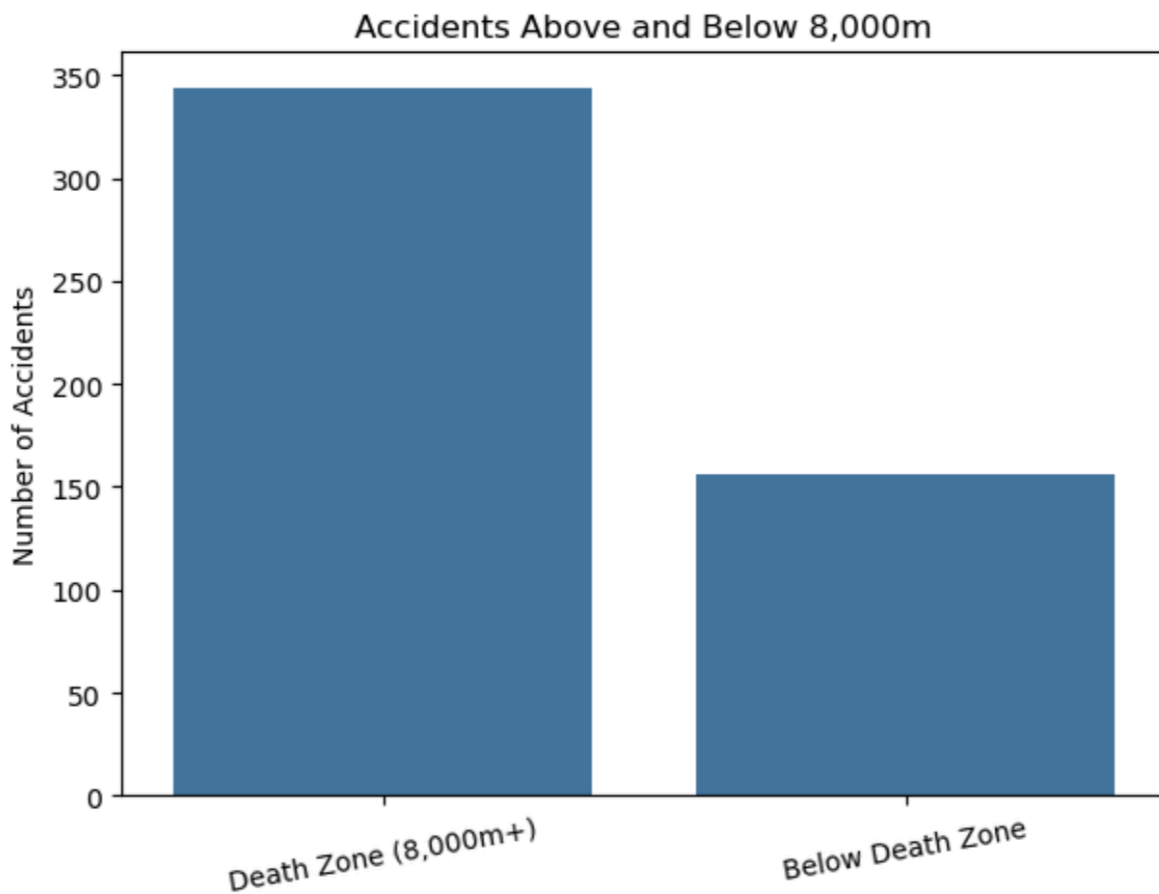
Only 31.2% of records occurred below 8,000 metres.

This makes extreme altitude the strongest pattern identified in the dataset.

8. Death Zone Analysis

The dataset was divided into two groups:

Zone	Reco rds	Perce ntage
Death Zone (8,000m+)	344	68.8%
Below Death Zone	156	31.2%



Finding

More than two-thirds of the recorded incidents occurred at or above 8,000 metres.

This finding is consistent with the analysis showing that altitude sickness was the leading cause of death.

9. Cause of Death at 8,000m+

A deeper analysis was performed to identify the leading causes specifically within the 8,000m+ altitude group.

Cause	Percentage of 8,000m+ Records
Altitude Sickness	32.6%
Exhaustio n	20.3%
Hypother mia	8.7%
Falling	8.1%
Avalanche	5.5%

Key Finding

Altitude sickness becomes particularly prominent at extreme altitude.

Within the 8,000m+ group, approximately one-third of recorded incidents were associated with altitude sickness.

This is one of the most important relationships identified by the analysis.

10. Seasonal Analysis

Season	Recorded Accidents	Percentage
Summer	193	38.6%
Winter	128	25.6%
Spring	91	18.2%
Autumn	88	17.6%

Finding

Summer had the highest number of recorded accidents, with 193 records.

Winter followed with 128 records.

Important Interpretation

The higher number of summer records does not necessarily mean that summer has a higher accident risk.

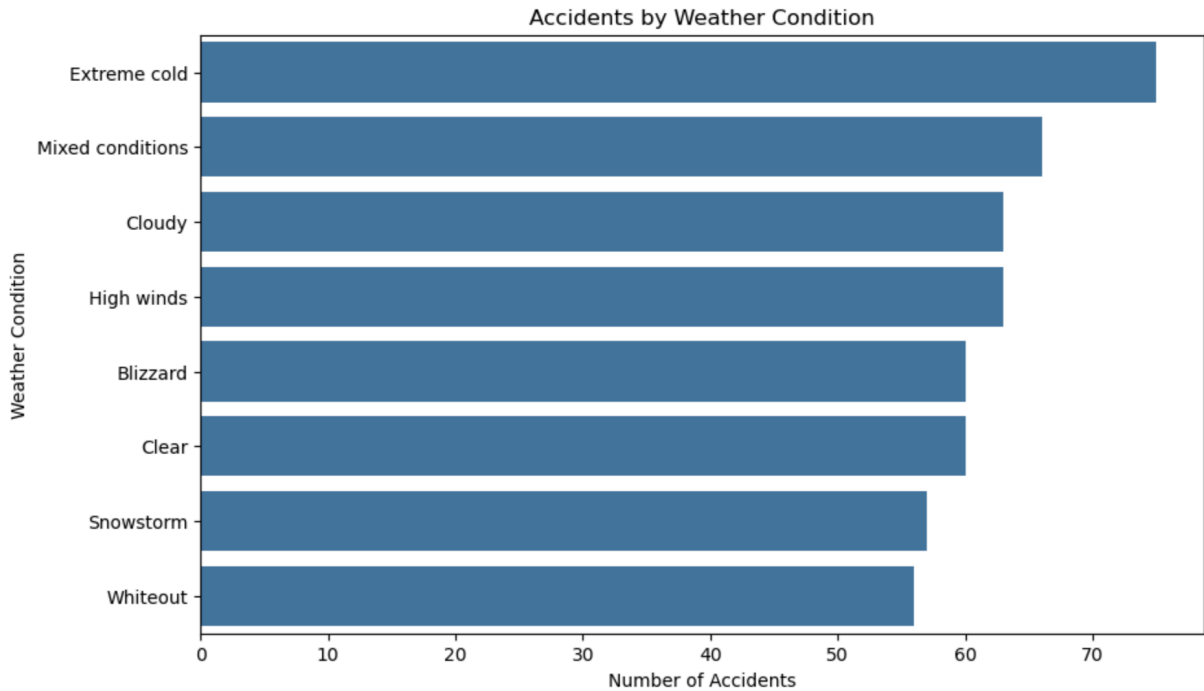
The dataset does not include:

- Number of climbers by season
- Number of expeditions by season
- Time spent climbing
- Route exposure

Therefore, these figures should be treated as recorded accident counts.

11. Weather Conditions

Weather Condition	Reco rds
Extreme cold	75
Mixed conditions	66
High winds	63
Cloudy	63
Clear	60
Blizzard	60
Snowstorm	57
Whiteout	56



Finding

Extreme cold was the most frequently recorded weather condition, with 75 records.

However, weather conditions should also be interpreted carefully because the dataset does not tell us how frequently each weather condition occurred during expeditions.

12. Experience Level

Experience	Reco
Level	rds

Novice	94
Experience d	86
Expert	85
Intermediat e	85
Sherpa	76
Profession al Guide	74

Finding

Novice climbers had the highest recorded accident count at 94 records.

However, the difference between the highest and lowest categories was only 20 records.

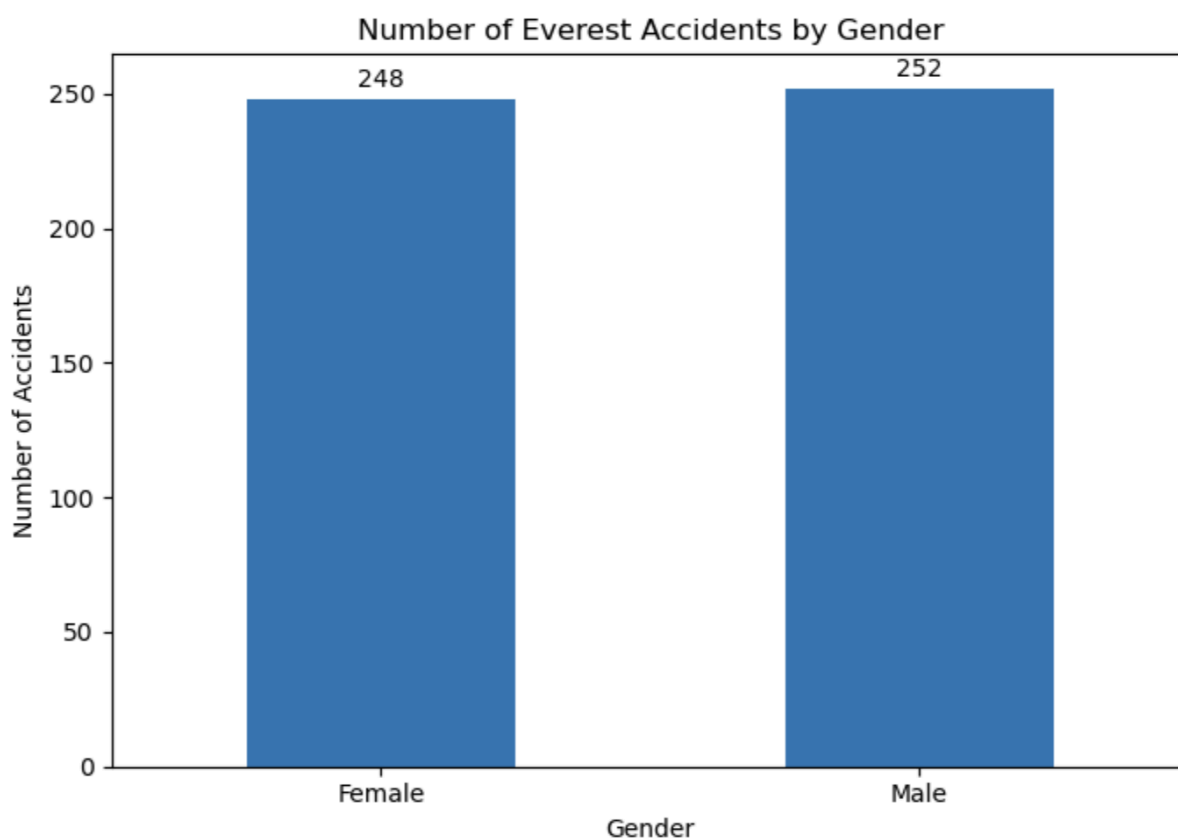
Therefore, the analysis does not support the conclusion that experience level alone is responsible for the majority of recorded accidents.

13. Gender Analysis

Gender	Records	Percentage
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Male	252	50.4%
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Female	248	49.6%
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Finding

The dataset shows an almost equal distribution between male and female records.

The difference is only four records.

This suggests that gender is relatively balanced within this dataset.

Again, this does not represent gender-specific risk because the dataset does not provide the underlying number of male and female climbers.

14. Age Analysis

The recorded ages ranged from 20 to 69 years.

Statistic	Age
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Mean	36.3
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Median	31
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Minimum	20
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Maximum	69
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Age Groups

Age Group	Records
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20–29	184
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30–39	176
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40–49	44
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50–59	53
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60+	43
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Finding

The majority of recorded individuals were between 20 and 39 years old.

Combined:

20–39 = 360 records

This represents 72% of the dataset.

15. Age and Altitude Relationship

The correlation between age and recorded altitude was approximately:

0.01

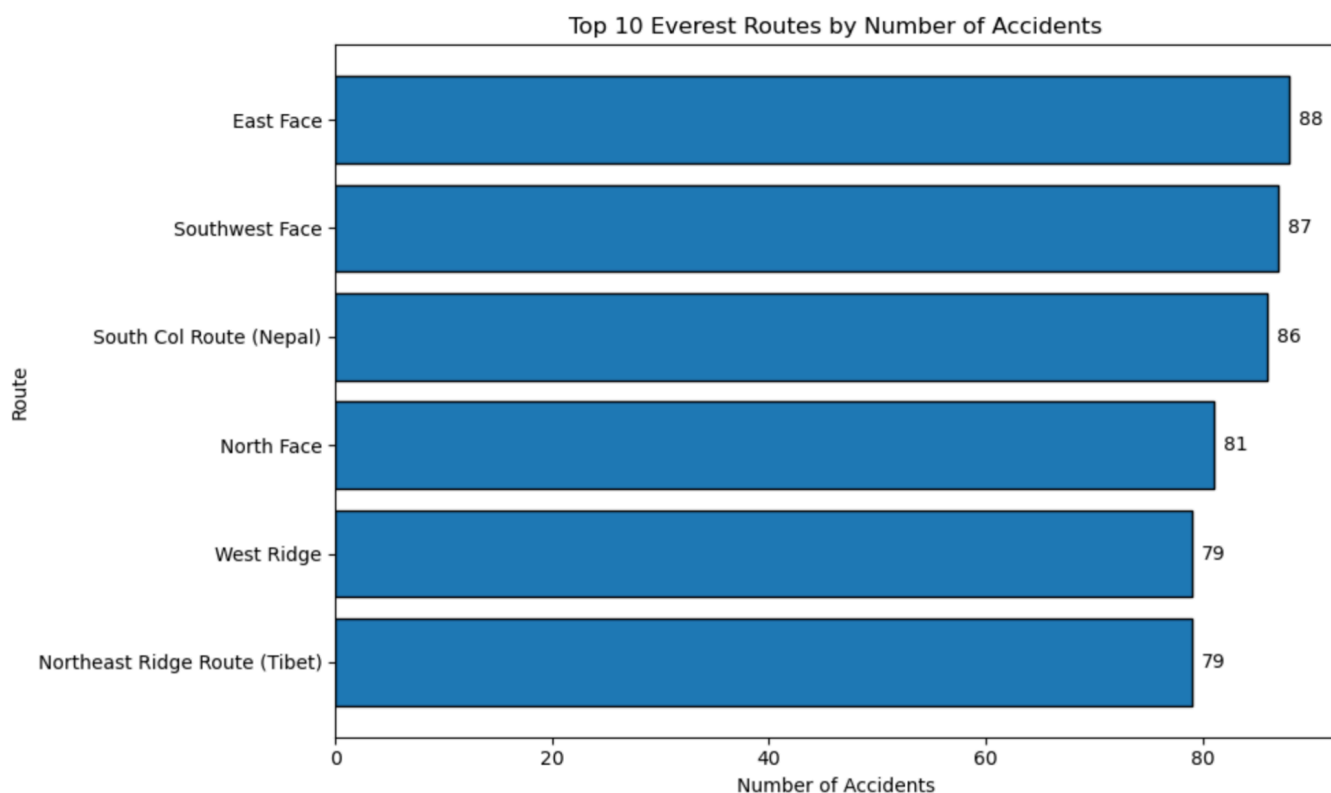
This indicates essentially no linear relationship between age and altitude in this dataset.

Interpretation

Older individuals were not systematically recorded at substantially higher or lower altitudes than younger individuals.

16. Route Analysis

Route	Records
East Face	88
Southwest Face	87
South Col Route (Nepal)	86
North Face	81
West Ridge	79
Northeast Ridge Route (Tibet)	79



Finding

The East Face had the highest recorded accident count at 88.

However, the range between routes was relatively narrow, with all routes recording between 79 and 88 cases.

Important Limitation

These figures cannot be used to determine which route is most dangerous because the dataset does not provide the number of climbers using each route.

17. Nationality Analysis

The dataset contains records from 30 nationalities.

The highest recorded counts included:

Nationality	Recorded Accidents
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Switzerland	31
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South Africa	26
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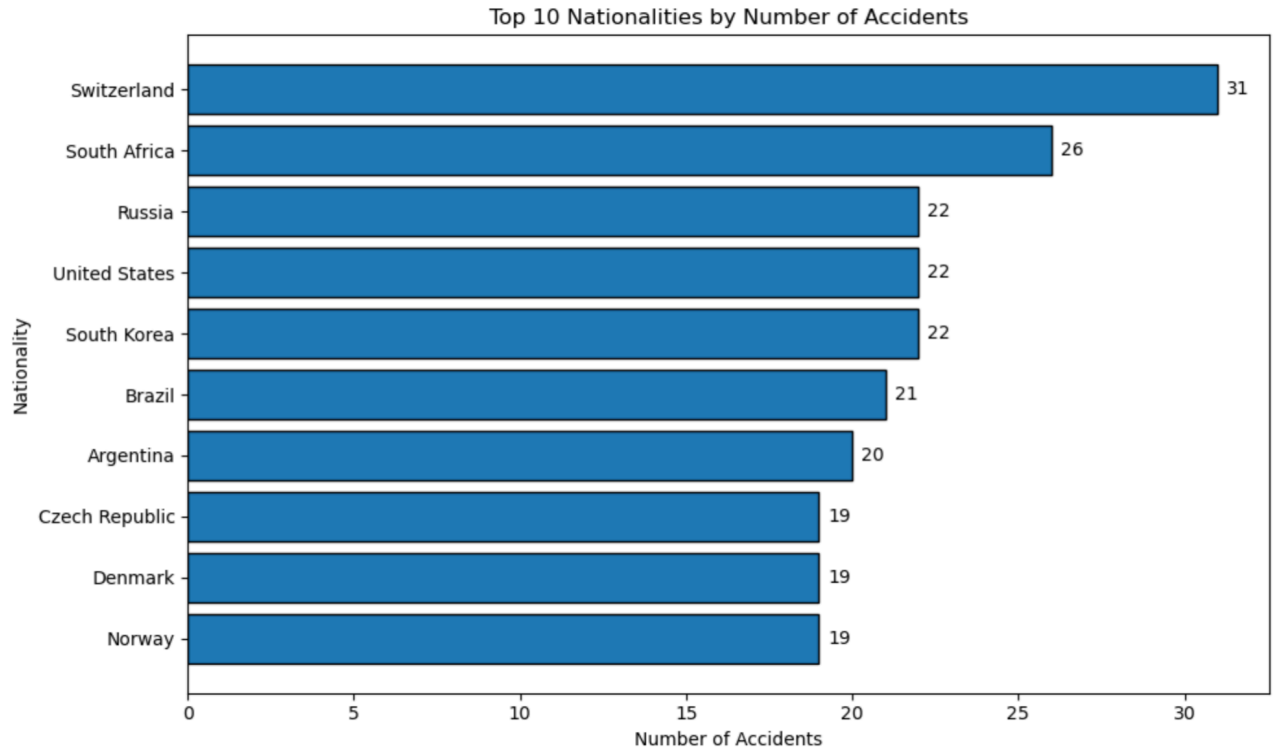
United States	22
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South Korea	22
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Russia	22
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Brazil	21
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Argentina	20
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Interpretation

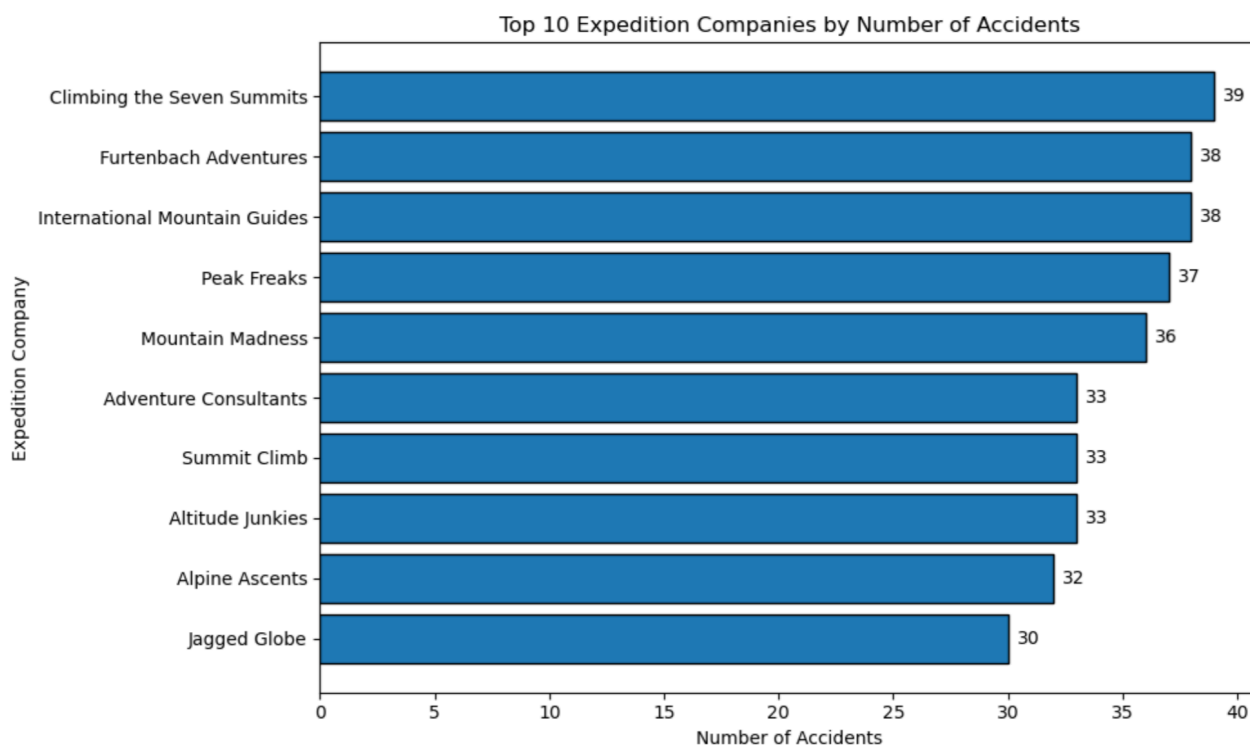
Switzerland had the highest recorded count.

18. Expedition Company Analysis

The dataset contains records associated with 16 expedition companies.

The highest recorded counts included:

Expedition Company	Reco rds
Climbing the Seven Summits	39
International Mountain Guides	38
Furtenbach Adventures	38
Peak Freaks	37
Mountain Madness	36
Adventure Consultants	33
Altitude Junkies	33
Summit Climb	33



Interpretation

These figures represent the number of records in the dataset associated with each company.

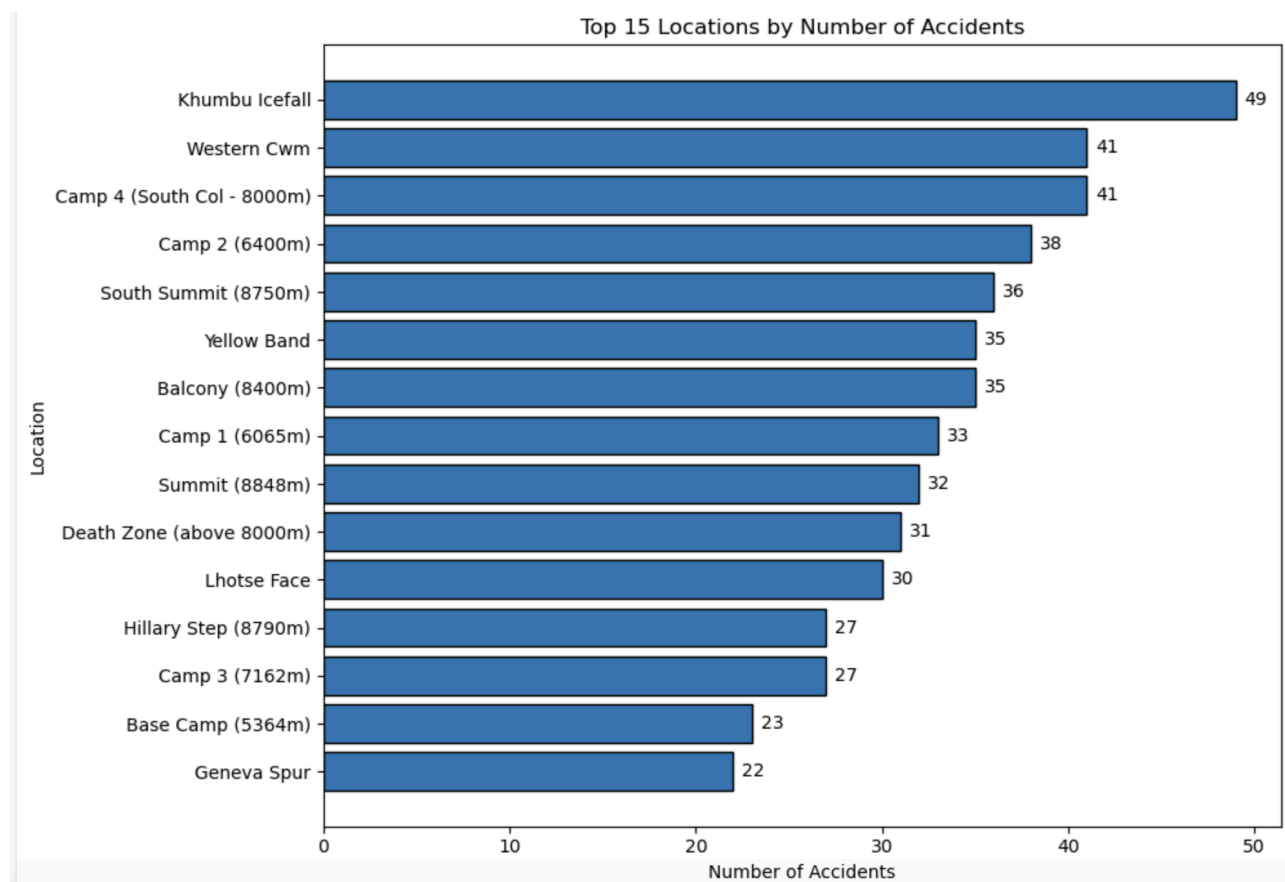
They should not be interpreted as evidence that one expedition company has a higher accident rate than another.

A meaningful comparison would require the number of clients or expeditions managed by each company.

19. Location Analysis

The most frequently recorded locations included:

Location	Records
Khumbu Icefall	49
Camp 4 (South Col - 8000m)	41
Western Cwm	41
Camp 2 (6400m)	38
South Summit (8750m)	36
Balcony (8400m)	35
Yellow Band	35
Camp 1 (6065m)	33
Summit (8848m)	32



Finding

The Khumbu Icefall had the highest recorded number of incidents at 49 records.

20. Deep-Dive Finding: Altitude and Cause

The cross-analysis of altitude and cause of death produced one of the strongest insights.

At 8,000m+, the leading causes were:

1. Altitude sickness — 32.6%
2. Exhaustion — 20.3%
3. Hypothermia — 8.7%
4. Falling — 8.1%
5. Avalanche — 5.5%

Interpretation

The concentration of altitude sickness and exhaustion at extreme altitude suggests that physiological challenges associated with very high altitude are an important feature of the recorded incidents.

This is a pattern in the dataset and should not be interpreted as a causal conclusion.

21. Key Findings Summary

The most important findings from the analysis are:

Finding 1 — Extreme altitude dominates

68.8% of recorded accidents occurred at 8,000m or above.

Finding 2 — Altitude sickness is the leading cause

Altitude sickness accounted for 25.8% of all records.

Finding 3 — Two causes account for a substantial proportion

Altitude sickness and exhaustion together represented 42.0% of all records.

Finding 4 — Recorded accidents declined after 2021

Recorded incidents fell from 93 in 2021 to 70 in 2025, a reduction of approximately 24.7%.

Finding 5 — Summer has the highest recorded count

Summer represented 193 records or 38.6% of the dataset.

Finding 6 — Extreme cold was the leading weather category

Extreme cold was associated with 75 records.

Finding 7 — Novices had the highest experience-category count

Novice climbers represented 94 records.

Finding 8 — Gender distribution was balanced

Male records represented 50.4% and female records 49.6%.

Finding 9 — Most records involved younger adults

Individuals aged 20–39 represented 72% of the dataset.

Finding 10 — Age and altitude showed almost no linear relationship

The correlation was approximately 0.01.

25. Conclusion

The analysis of 500 Mount Everest accident records from 2020–2025 identified several strong patterns.

The most significant finding was the concentration of recorded incidents at extreme altitude. 344 records, representing 68.8% of the dataset, occurred at 8,000 metres or above.

Altitude sickness was the most frequently recorded cause of death, accounting for 129 records or 25.8%. Exhaustion was the second-largest cause at 81 records. Together, these two categories represented 42% of the dataset.

Recorded accident counts peaked in 2021 and subsequently declined through 2025. Summer recorded the highest number of incidents, while extreme cold was the most common weather category.

The analysis also found an almost equal gender distribution and a strong concentration of records among individuals aged 20–39.

However, the most important analytical limitation is the absence of exposure data. Without knowing how many people climbed, which routes they used, or how many expeditions each company conducted, the analysis cannot determine actual accident rates or risk.
